



A Fully Automatic Pipeline of Identification, Segmentation, and Subtyping of Aortic Dissection from CT Angiography

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Abstract

Purpose Aortic dissection (AD) is a rare condition with a high mortality rate, necessitating accurate and rapid diagnosis. This study develops an automated deep learning pipeline for identifying, segmenting, and Stanford subtyping AD using computed tomography angiography (CTA) images.

Methods This pipeline consists of four interconnected modules: aorta segmentation, AD identification, true lumen (TL) and false lumen (FL) segmentation, and Stanford subtyping. In the aorta segmentation module, a 3D full-resolution nnU-Net is trained. The segmented aorta's boundary is extracted using morphological operations and projected from multiple views in the AD identification module. AD identification is then performed using the multi-view projection data. For AD cases, a 3D nnU-Net is further trained for TL/FL segmentation based on the segmented aorta. Finally, a network is trained for Stanford subtyping using multi-view maximum density projections of the segmented TL/FL. A total of 386 CTA scans were collected for training, validation, and testing of the pipeline.

Results For AD identification, the method achieved an accuracy of 0.979. The TL/FL segmentation for TypeA-AD and Type-B-AD achieved average Dice coefficient of 0.968 for TL and 0.971 for FL. For Stanford subtyping, the multi-view method achieved an accuracy of 0.990.

Conclusion The automated pipeline enables rapid and accurate identification, segmentation, and Stanford subtyping of AD using CTA images, potentially accelerating the diagnosis and treatment. The segmented aorta and TL/FL can also serve as references for physicians. The code, models, and pipeline are publicly available at <https://github.com/zhuangCJ/A-pipeline-of-AD.git>.

Keywords Aortic dissection · Identification · Segmentation · Stanford subtyping · Deep learning · Computer tomography angiography

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Introduction

Aortic dissection (AD) is a disease caused by the tearing of the intima of the aorta or internal bleeding within the aortic wall, leading to the separation of the layers of the aortic wall and the formation of a dissection flap. This flap creates a true lumen (TL) and a false lumen (FL) on either side. AD can be classified into two types according to the Stanford subtyping system. Stanford Type A (TAAD) refers to dissection involving the ascending aorta, while Stanford Type B (TBAD) indicates dissection confined to the descending aorta [1]. The incidence of aortic dissection is approximately 3 per 100,000, with patients typically over 50 years old, and more than 60% of cases occurring in males. The short-term mortality rate for untreated aortic dissection is over 40% [2, 3].

The primary diagnostic imaging modalities for AD include computed tomography (CT) and magnetic resonance imaging (MRI) [4]. Treatment options for AD encompass open surgery, endovascular repair, and medical therapy [3, 4], with a specific approach determined by the patient's condition. The low incidence of AD and its complex clinical presentation contribute to a lack of awareness among non-cardiovascular specialists and general clinicians, leading to misdiagnosis or missed diagnoses. Furthermore, the high mortality rate, which exceeds 40% for untreated cases, highlights the critical need for rapid diagnosis and assessment. Therefore, the rapid and automated identification of AD, along with segmentation of the aorta and TL/FL, is of significant clinical importance.

Computed tomography angiography (CTA) provides detailed vascular imaging with high spatial resolution through multi-row spiral or dual-source CT. Unlike traditional angiography, CTA is less invasive, as the contrast agent is injected via the forearm veins, minimizing trauma. The iodine-based contrast enhances the distinction between blood vessels and surrounding tissue, aiding in the detection of both anatomical and pathological changes. Faster, cost-effective, and capable of imaging a broader range of vascular systems, CTA is often preferred over traditional digital subtraction angiography [5]. CTA is commonly used for detecting cerebrovascular, cerebral or coronary artery, and aortic-related diseases, and is one of the primary methods for detecting aortic dissection [6, 7].

Deep learning (DL) has shown significant potential in feature extraction and representation, especially with the advent of convolutional neural networks (CNN) [8] and U-Net [9]. Additionally, DL has gained prominence in aorta segmentation, TL/FL segmentation, AD identification, and Stanford subtyping.

For aorta and TL/FL segmentation, several DL methods have been proposed. Sieren et al. [10] used a 3D U-Net for

aortic segmentation. Aghilinejad et al. [11] applied three DL frameworks to segment the aorta at the slice level. Feng et al. [12] developed a two-step segmentation network for thrombotic aorta in postoperative patients. Xiang et al. [13] integrated a flap attention module for aorta segmentation. Lin et al. [14] proposed a deformable constraint transport network for aortic segmentation. Kesävuori et al. [15] used V-Net to extract the aortic outer surface in TBAD patients. Chen et al. [16] employed two cascaded neural networks and a straightening-based approach for aortic segmentation in TBAD. Zhao et al. [17] enhanced segmentation accuracy using deep mesh deformation. Cao et al. [18] used cascaded 3D U-Net models for aorta and TL/FL segmentation. Zhang et al. [19] introduced a global-local fusion learning mechanism for aorta and TL/FL in TBAD. Li et al. [20] developed a 2D cascaded CNN for stepwise segmentation of the aorta and TL contours in cross-sectional images. Yu et al. [21] used a 3D U-Net and intersection over union tracing for TL/FL localization in TBAD. Zhang et al. [22] proposed a method for segmenting the TL/FL and aortic branches in TBAD, adapting to small-sample learning. Yao et al. [23] introduced the ImageTBAD dataset and baseline models. Feiger et al. [24] proposed a variant of the 2D U-Net for segmenting the aorta, TL, and FL in TBAD.

For AD identification and Stanford subtyping, Cheng et al. [25] used a 2D U-Net to segment CTA data in the slice plane and performed circularity analysis to detect dissection, followed by patient-level voting. Huang et al. [26] applied two cascaded CNNs for AD identification. Dong et al. [27] utilized YOLOv5 with a Convolutional Block Attention Module (CBAM) for slice-level AD recognition. Hata et al. [28] proposed an AD detection method using non-contrast-enhanced CT. Tan et al. [29] combined morphology and deep learning for AD detection. Harris et al. [30] developed a method for detecting aortic injuries, including dissection and rupture. Huang et al. [31] proposed a two-step neural network to classify Negative, TAAD, and TBAD. Xu et al. [32] used Mask R-CNN for segmenting the ascending and descending aorta and aortic arch, employing two ResNet-18 models to detect dissections in the ascending aorta and aortic arch, and determining disease status based on the number of affected slices.

Despite these advances, most studies focus on single aspects of the diagnostic process, often exploring specific tasks in isolation. Additionally, they fail to fully leverage prior segmentation knowledge for classification tasks, limiting comprehensive diagnostics. Such approaches do not replicate the integrative, multi-dimensional analysis typical of radiologists. To bridge this gap, we introduce a novel pipeline that emulates a holistic diagnostic approach, combining multiple deep learning (DL) tasks into a unified pipeline.

In this study, inspired by the diagnostic procedure of radiologists, we designed an automated pipeline for AD diagnosis and subtyping using CTA images. The pipeline includes aorta segmentation, AD identification, TL/FL segmentation, and Stanford subtyping. Key contributions are: (1) an automated pipeline that performs well in diagnosing and subtyping AD on an in-house dataset, with four interconnected modules: aortic segmentation, AD identification, TL/FL segmentation, and Stanford typing; (2) accurate segmentation of the aorta and true/false lumens for both TAAD and TBAD, with segmentation knowledge enhancing classification tasks; (3) a multi-view projection method for patient-level prediction (identification and subtyping) using segmented regions, without relying on manually set thresholds. The code, models, and pipeline are publicly available at <https://github.com/zhuangCJ/A-pipeline-of-AD.git>. Clinicians can also contact the corresponding author to request the segmented results if running the pipeline is inconvenient.

Materials and Methods

Dataset

This study utilized three datasets: one in-house dataset and two publicly available datasets, namely the TBAD dataset [23] and AVT (Aortic Vessel Tree, AVT) dataset [33, 34]. These datasets were applied to one or more of the four tasks/modules: aorta segmentation, TL/FL segmentation, AD identification, and Stanford subtyping. In the following sections, we first describe each of the three datasets in details, followed by an explanation of how these datasets were divided into subsets and used across the different modules.

In-house Dataset

The clinical data and CTA images of 412 participants were collected from the General Hospital of Northern Theater Command between March 2019 and July 2023. There were 151 TAAD patients, 156 TBAD patients, and 105 participants

without AD (non-AD). Data were filtered according to the following criteria: (1) exclusion of cases with multiple layers of rupture in the aortic region; (2) exclusion of cases with thrombus in the aorta; (3) exclusion of cases with significant motion artifacts. After applying these exclusions, the dataset retained 137 cases of TAAD, 148 cases of TBAD, and 101 non-AD cases. All AD diagnoses were confirmed by experienced radiologists with expertise in cardiovascular imaging: YS (6 years), JH (7 years), and BY (over 30 years). This study was approved by the Medical Ethics Committee of the General Hospital of Northern Theater Command. Informed consent was waived due to the retrospective nature of the study, which made obtaining consent impractical. Additionally, necessary data de-identification procedures were implemented to ensure confidentiality.

Detailed information on the clinical data of participants and CTA scanning parameters is provided in Table 1. The AD group has a significantly higher male-to-female ratio and age compared to the Non-AD group ($p < 0.005$). The male-to-female ratio in the Stanford B group is significantly higher than that in the Stanford A group ($p < 0.005$). No significant differences are observed for the other parameters.

For the in-house dataset, it is initially divided into two subsets: the labeled ($n=202$; including 101 non-AD, 50 TAAD, and 51 TBAD) and unlabeled ($n=184$). Not all of the cases were labelled because the labeling is time-consuming and 202 labeled cases have been proven to be sufficient for the segmentation modules. Semi-automatic labeling was performed on 202 CTA images (101 Non-AD, 50 TAAD, and 51 TBAD) using Mimics (Materialise Corp., Belgium), under the guidance of experienced radiologists YS, JH, and BY. The labeled regions included the ascending aorta, aortic arch, and descending aorta, excluding the innominate artery, left carotid artery, left subclavian artery, and iliac arteries branching from the descending aorta. The foreground pixels in the labeled regions were assigned a value of 1, while the background pixels were assigned a value of 0. The labels were saved in NIFTI format.

Table 1 Characteristics of participants and CTA image acquisition parameters

Parameter	Participant			AD		
	AD	Non-AD	<i>p</i> value	Stanford A	Stanford B	<i>p</i> value
Gender (male/female)	214/71	68/33	$<0.001^a$	95/42	119/29	$<0.001^a$
Age (years)	56.4 ± 12.7	48.7 ± 13.2	$<0.001^b$	55.6 ± 12.1	57.1 ± 13.3	0.3039 ^b
Slice thickness (mm)	0.625	0.625	-	0.625	0.625	-
kVp (kV)	120	120	-	120	120	-
Pixel size (mm)	0.819 ± 0.103	0.838 ± 0.092	0.0399 ^b	0.820 ± 0.082	0.818 ± 0.078	0.7574 ^b
X-ray tube current (mA)	334.3 ± 133.7	340.9 ± 132.6	0.6576 ^b	334.8 ± 137.4	333.7 ± 130.0	0.9408 ^b
Exposure (mA*s)	$87.2.7 \pm 113.9$	95.9 ± 113.2	0.5010 ^b	83.5 ± 113.3	90.9 ± 114.4	0.5687 ^b

^a represents the chi-square test, and ^b represents the two-sample t-test.

TBAD Dataset

This dataset consisted of 3D CTA images collected from Guangdong Provincial People's Hospital between January 1, 2013, to April 23, 2019, totaling 100 scans. The images were acquired using various scanners (GE Healthcare, Siemens, Philips), leading to substantial variations in voxel size, resolution, and imaging quality. All images were pre-operative CTA scans of TBAD cases, with the axial view capturing the vessels around the neck and the arms. In this dataset, segmentation labeling was conducted by a team of two cardiovascular radiologists, who marked regions including the true lumen, false lumen, and false lumen thrombus [23]. In the present study, no additional annotations were generated; all experiments were conducted using the original labels provided with the dataset.

AVT Dataset

The AVT dataset included 56 multicenter aortic CTA scans, covering the aortic arch and its branches, as well as the abdominal aorta and iliac arteries. The labeled regions encompass the aorta and its branches.

Dataset Subdivision and Usage

The details of dataset subdivision and usage are shown in Fig. 1. For the aorta segmentation module, all 50 cases in the AVT dataset are used (blue Line (1) with an arrow in Fig. 1) as the external validation. The AVT dataset has been previously used for aortic segmentation in other studies [33, 34]. All 100 cases in the TBAD dataset are also used as the external validation (blue Line (2) with an arrow in Fig. 1).

Here, it is noted that the external validation indicates to validate the model's generalization capability by using the external datasets which are not used for tuning hyperparameters, adjusting fit, etc. For the in-house dataset, it is initially divided into two subsets: the labeled ($n=202$; including 101 non-AD, 50 TAAD, and 51 TBAD) and unlabeled ($n=184$). The labeled subset is used for the aorta segmentation module, as indicated by the blue Line (3) with an arrow in Fig. 1. It is important to note that we did not include Non-AD data in the training and validation sets, as we observed that adding a large amount of Non-AD data tended to bias the network's segmentation results, leading to the neglect of subtle structures in the dissection. Therefore, the 101 labeled AD cases in this subset are further randomly divided into three parts for training, validation, and testing, following a 4:1:1 ratio. Meanwhile, all 101 labeled Non-AD cases in this subset are assigned to the testing set. Specifically, training is conducted on 68 cases (34 TAAD and 34 TBAD), validation is performed on 17 cases (8 TAAD and 9 TBAD), and testing involves 117 cases, consisting of 8 TAAD, 8 TBAD, and 101 Non-AD cases.

For the TL/FL segmentation module, all 100 cases in the TBAD dataset are used as the external validation (orange Line (4) with an arrow in Fig. 1). In the in-house dataset, the 101 labeled AD cases containing TL and FL are further used for the TL/FL segmentation module, as indicated by the orange Line (5) with an arrow in Fig. 1. These cases are divided into training ($n=68$; 34 TAAD, validation ($n=17$; 8 TAAD), and testing ($n=16$; 8 TAAD) subsets.

For the AD identification module, all 386 cases in the in-house dataset are used (purple Line (6) with an arrow in Fig. 1). A five-fold cross-validation approach is applied for training, validation, and testing with a ratio of 3:1:1. It

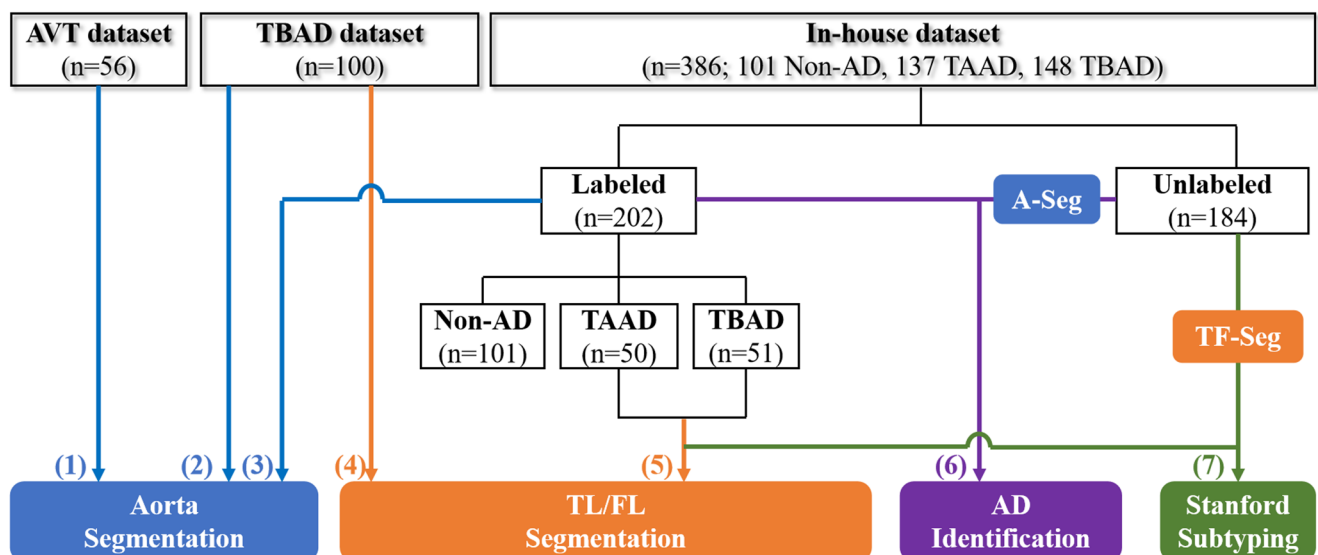


Fig. 1 The three datasets, the subdivision of the in-house dataset, and their usage to four modules. A-Seg and TF-Seg indicate the trained aorta segmentation network and TL/FL segmentation network, respectively

is noted that the 184 unlabeled cases must be input into the developed aorta segmentation network to obtain the aorta mask before AD identification.

For the Stanford subtyping module, the 101 labeled AD cases and the 184 unlabeled AD cases in the in-house dataset are used (green Line (7) with an arrow in Fig. 1), with five-fold cross-validation applied in the same manner. The 184 unlabeled cases must be input into the developed TL/FL segmentation network to obtain the masks before Stanford subtyping.

Although the “Non-AD (n=101)” box is not directly connected to any module, the 101 Non-AD cases contribute to both the aorta segmentation and AD identification modules. As illustrated in Fig. 1, the “Non-AD (n=101)” box is linked to these two modules via its direct parent, i.e., the “Labeled (n=202)” box.

Overview of Study Procedure

As shown in Fig. 2, the proposed pipeline automatically identifies, segments, and classifies aortic dissection using CTA images. It consists of four modules: (1) aorta segmentation, (2) aortic dissection identification, (3) true and false lumen segmentation, and (4) Stanford subtyping. The following sections describe the details of each module.

Aorta Segmentation

Data preprocessing is essential due to the data being acquired from different CT scanners. First, the CT values of the raw data were normalized within a fixed window (window width=4095 HU, window level=1024 HU). Next, the CT images were interpolated to a size of $512 \times 512 \times N$ using cubic spline interpolation to match the network input requirements. During training, the voxel spacing of the CT images was normalized to the mean spacing calculated across all training CT scans, and the images were further partitioned into patches of $96 \times 96 \times 96$ for model training. Finally, the CTA and label data were cropped based on the number of foreground pixels in the labels, removing transverse slices with fewer than 200 foreground pixels to exclude slices with insufficient information at the boundaries. Additionally, the data from the external validation datasets were preprocessed by standardizing the window width and level to match our dataset.

Aortic segmentation was performed on the processed data using the 3D nnU-Net. nnU-Net, introduced by Isensee et al. in 2018, is an adaptive segmentation network based on U-Net, which has demonstrated strong performance in various medical image segmentation tasks [35]. nnU-Net employs a patch strategy to handle medical images of varying sizes by subdividing large images into smaller blocks or patches. Segmentation predictions are made for each patch, and the resulting segmentations are then combined

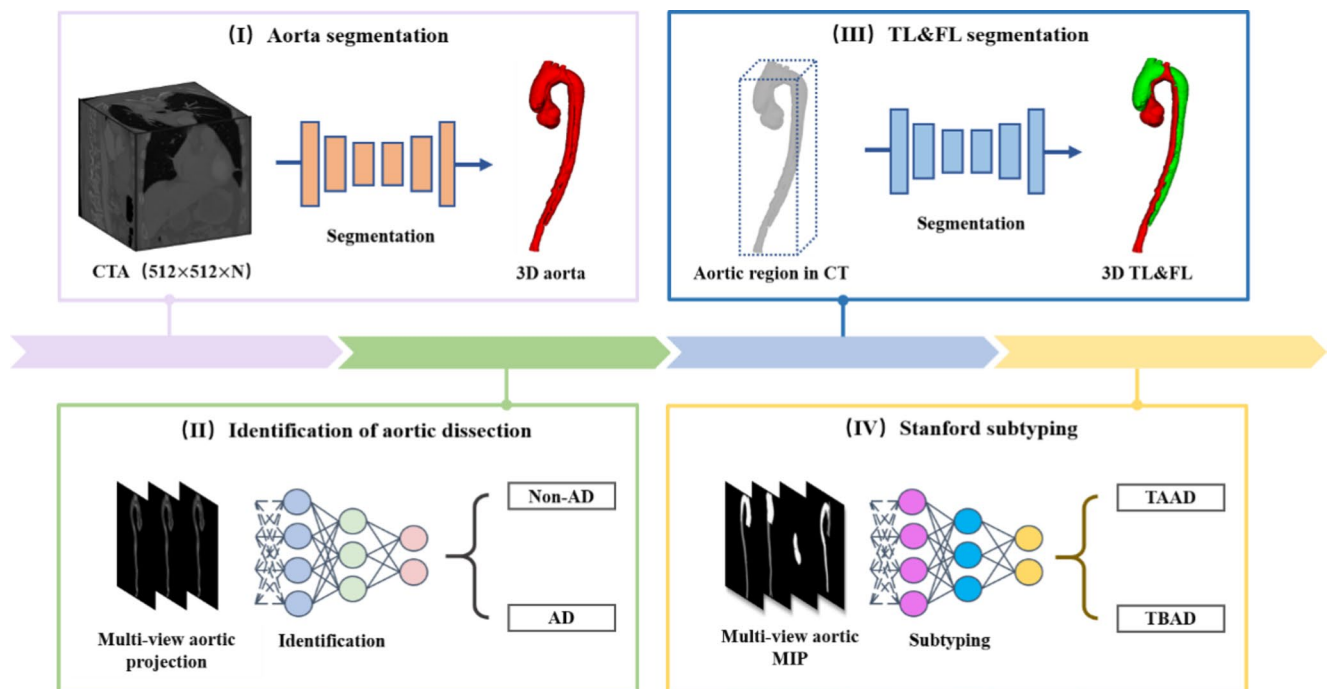


Fig. 2 The pipeline of AD identification, segmentation, and Stanford subtyping

to generate the full image segmentation. In this module, we utilized the full-resolution 3D nnU-Net for aortic segmentation.

The network is trained for 1000 epochs, with an early stopping mechanism implemented: training halts if the network does not converge within 30 consecutive epochs. The loss function is the average of Dice loss and cross-entropy loss. The formulas for Dice loss and cross-entropy loss are as follows:

$$L_{Dice} = -\frac{2}{K} \sum_{k \in K} \frac{\sum_{i \in I} u_i^k v_i^k}{\sum_{i \in I} u_i^k + \sum_{i \in I} v_i^k} \quad (1)$$

$$L_{CE} = -\frac{1}{I} \sum_{k \in K} \sum_{i \in I} u_i^k \cdot \log(v_i^k) \quad (2)$$

where K represents the number of segmentation classes, I denotes the pixel index, u_i^k represents the predicted probability of the k -th class for the i -th pixel, and v_i^k is the ground truth value for the k -th class of the i -th pixel.

Stochastic Gradient Descent (SGD) is used as the optimizer, with an initial learning rate of 0.01 and a momentum set to 0.99. The learning rate follows a decay formula as the epochs progress, which is given by:

$$lr = 0.01 \times \left(1 - \frac{epoch}{1000}\right)^{0.9} \quad (3)$$

To improve segmentation performance, we applied post-processing by performing the maximum connected component operation on the segmentation results. A 26-connectivity kernel was used to retain the largest connected region, removing smaller isolated regions. This operation ensures that multiple regions are merged into a single, connected component. Since the aorta tears at a specific location, the true and false lumens are connected at some point, preventing complete separation. The 26-connectivity kernel was chosen to avoid disconnections caused by narrow linking areas and to ensure that as many voxels as possible are included in the connected component. This process is unaffected by image resolution, as all images were interpolated to $512 \times 512 \times N$ during the preprocessing stage. The post-processing step guarantees that only the largest connected

component remains, effectively eliminating small, isolated regions not connected to the main structure. It is important to note that this operation does not impact the segmentation of the aortic region, as its inherent connectivity is preserved.

We also compared other segmentation networks, including 3D U-Net [36], 3D ResU-Net, SwinU-Net [37], and nnFormer [38].

Identification of Aortic Dissection

We processed 386 masks, consisting of 202 labeled masks and 184 segmentation results. The processing procedure is as follows: (1) Exclude slices that do not contain the aorta from the 386 cases, (2) Use 3D morphological erosion and dilation operations to extract the aortic boundaries: perform 3D morphological erosion with a 1-pixel kernel and 3D dilation with a 1-pixel kernel on the aortic mask. Then, subtract the eroded mask from the dilated mask to obtain the aortic boundaries, (3) Project the boundaries: Accumulate the pixel values of the aortic boundaries in a specific direction. For example, in the coronal plane, perform an addition operation on all slice matrices along the coronal plane direction to create a new two-dimensional matrix, which is then resized to dimensions of 224×224 . The operational workflow is illustrated in Fig. 3. Through projection, we convert the three-dimensional data into two-dimensional data while retaining information about the dissection. The boundary extraction process enhances the representation of dissection information. To prevent blurring of dissection information due to the parallel alignment of the dissection with the projection plane, we chose to project onto the coronal plane, the coronal plane rotated 30 degrees to the right, and the coronal plane rotated 45 degrees to the right. Figure 4 illustrates the processing procedure and results for one case of AD and one case of non-AD.

As shown in Fig. 5, we constructed a classification network that takes three-view projection images as input. This network consists of two modules: the feature extraction module and the classification module. In the feature extraction module, we use ResNet-34 [39] combined with squeeze-and-excitation (SE) [40] block to extract features from three-view input, resulting in three corresponding feature vectors. These feature vectors are then passed to the

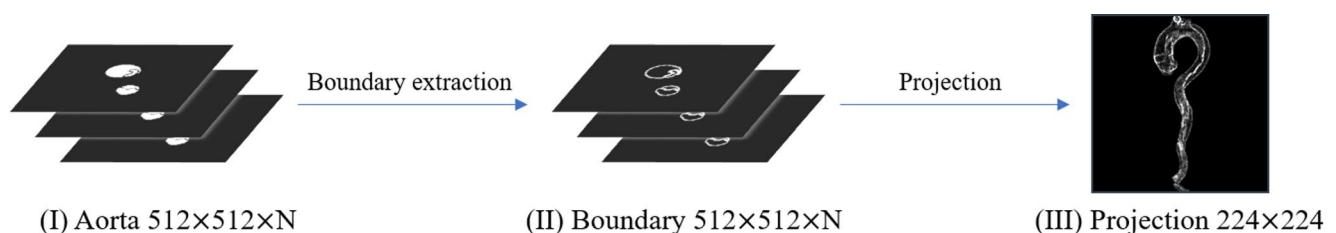


Fig. 3 Steps in data processing for the identification of aortic dissection

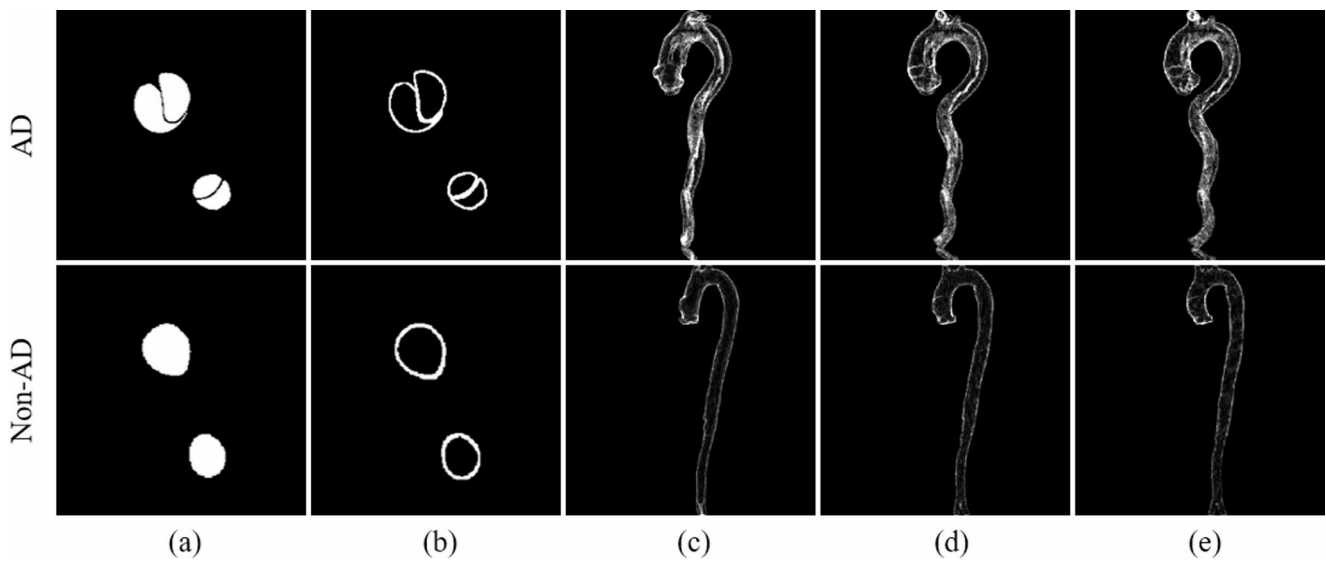


Fig. 4 Example of the processing procedure and results. (a) Aorta; (b) Aortic boundary; (c) Projection onto the coronal plane; (d) Projection onto a 30 degrees rotated coronal plane; (e) Projection onto a 45 degrees rotated coronal plane

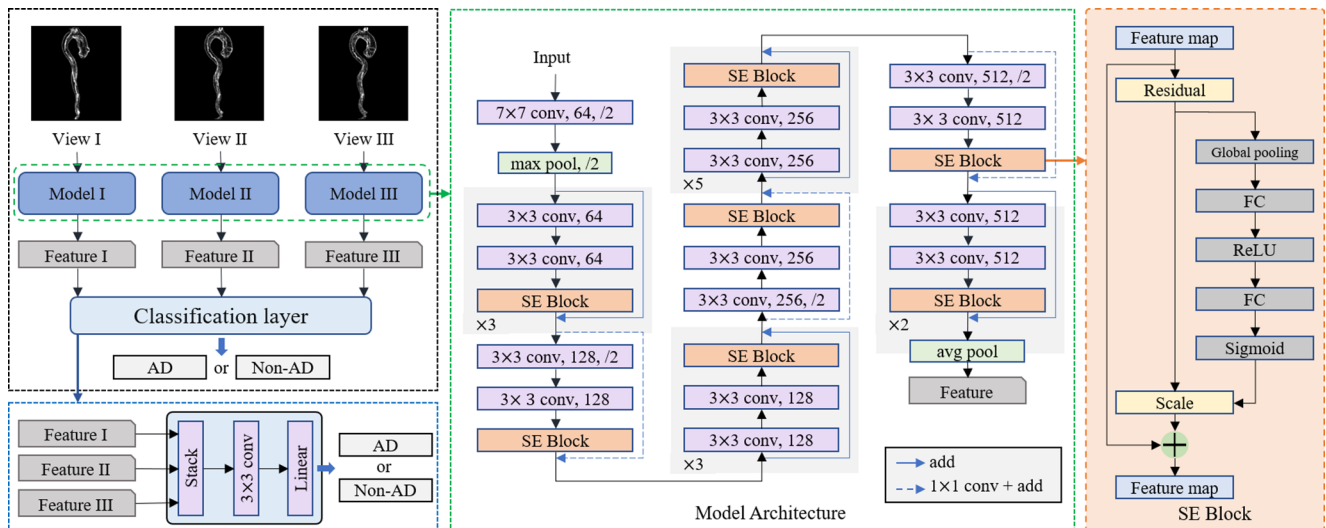


Fig. 5 Overview of classification method and network architecture

classification module. In the classification module, the feature vectors are merged into a feature matrix for fusion. The fused features are then passed through a convolutional layer before being fed into a fully connected layer. The output of the fully connected layer represents the prediction scores, which are used to calculate the loss function, and backpropagation is performed separately for both the feature extraction and classification modules.

We initialized the network with pre-trained parameters from ImageNet. The network was trained for 200 epochs with a batch size of 2. Early stopping was implemented, halting training if the validation loss did not improve for 30 consecutive epochs. Cross-Entropy Loss was used as the

loss function for the classification network, which is formulated as follows:

$$Loss_{CE} = -y_{true} \log(y_{pred}) - (1 - y_{true}) \log(1 - y_{pred}) \quad (4)$$

where y_{true} represents the true class (0 or 1) and y_{pred} represents the predicted class (0 or 1). AdamW is used as the optimizer, with an initial learning rate of 0.0001 and a weight decay of 0.05.

We compared the proposed method with other classification networks, including VGG16 [41], Inception-V3 [42], Vision Transformer (ViT) [43] and ResNet34 [39].

Segmentation of True Lumen and False Lumen

Using Mimics (Materialise Corp., Belgium), we further labeled the TL and FL. Under the guidance of expert radiologists, we annotated 50 cases of TAAD and 51 cases of TBAD. Background pixels were assigned a value of 0, TL foreground pixels were assigned a value of 1, and FL foreground pixels were assigned a value of 2. The aortic region was used as a mask to preserve the CT values within the masked area, while values outside the mask were set to 0. The processed data and labels were saved as NIFTI format files, with a size of $512 \times 512 \times N$. For the TBAD dataset, we preprocessed the data by standardizing the window width and level to match our dataset (window level=1024 HU, window width=4095 HU).

The full-resolution 3D nnU-Net was used for TL and FL segmentation. The network was trained for 1000 epochs, with an early stopping mechanism: training halts if the network does not converge for 30 consecutive epochs. The loss function is the average of Dice loss and cross-entropy loss. Stochastic Gradient Descent (SGD) was used as the optimizer, with an initial learning rate of 0.01 and a momentum of 0.99. We also compared the performance of our method with other segmentation networks, including 3D U-Net, 3D ResU-Net, SwinU-Net, and nnFormer.

Stanford Subtyping

A total of 285 masks were processed (137 TAAD and 148 TBAD), including 101 labeled masks and 184 TL/FL segmentation results. A crucial distinction between TAAD and TBAD is the position of the false lumen. Therefore, we performed maximum density projections on the masks in the coronal, sagittal, and transverse planes. In most cases, the position of the false lumen could be observed from these three planes. However, since aortic morphology varies among patients, projections in the coronal or sagittal planes sometimes show overlap between the ascending and descending aorta. To address this, we added a projection with a 30-degree right rotation in the coronal plane. With this rotation, the ascending and descending aorta no longer overlap in at least two of the planes: coronal (rotated 30 degrees), coronal, and sagittal. The maximum density projections of one case of TAAD and one case of TBAD are shown in Fig. 6.

In the Stanford subtyping task, we used a similar classification architecture as in the AD identification task, with the main difference being the increase in the number of input views from 3 to 4. The network was initialized with pre-trained parameters from ImageNet. Training was conducted for 200 epochs, with a batch size of 2. Early stopping was implemented, halting training if the validation loss did not decrease continuously for 30 epochs. The loss function used was Cross Entropy loss, and the optimizer was AdamW with an initial learning rate of 0.0001 and a weight decay

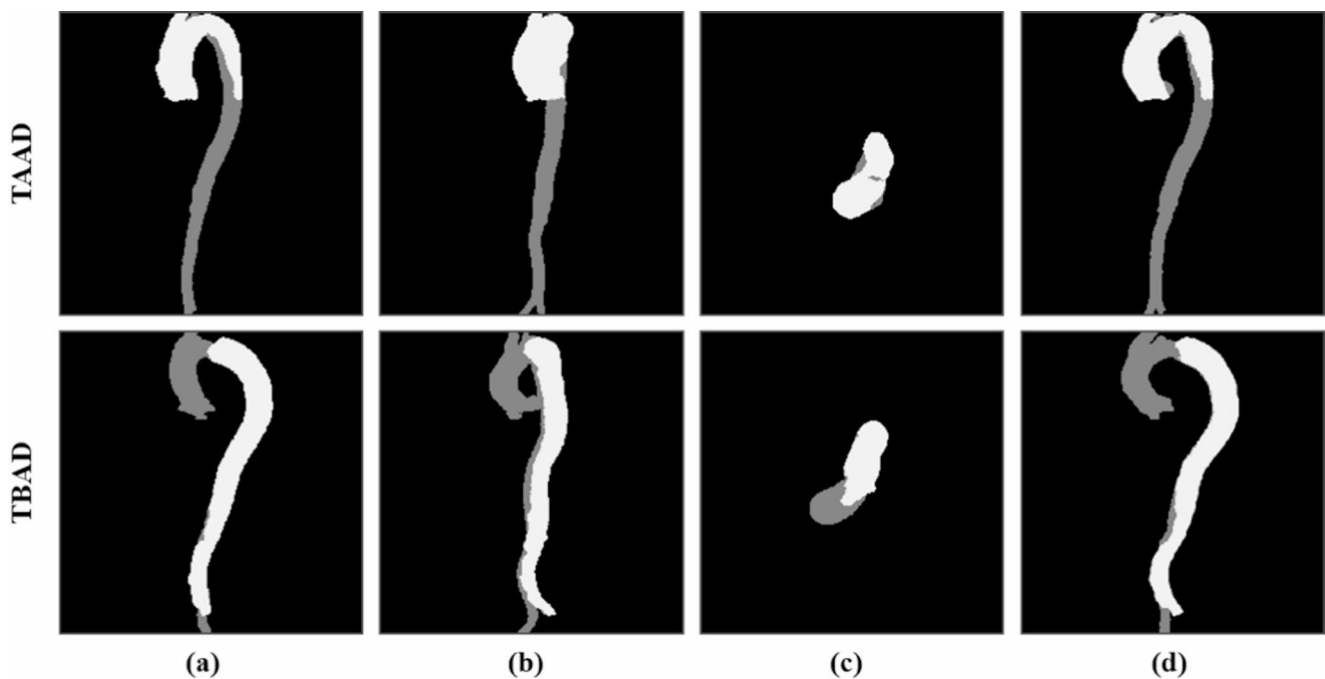


Fig. 6 Maximum density projections of TAAD and TBAD data from different angles. (a) Coronal plane; (b) Sagittal plane; (c) Transverse plane; (d) Coronal plane rotated to the right by 30 degrees

of 0.05. We compared our proposed method with other classification networks, including VGG16, Inception-V3, ViT, and ResNet-34.

Comparative Experiments

We conducted two comparative experiments. The first aimed to assess whether using the segmentation results of the aorta as the region of interest (ROI) could enhance the accuracy of TL/FL segmentation. The second experiment evaluated whether combining multiple views could improve the accuracy of AD identification and Stanford subtyping, compared to using a single-view projection.

Evaluation of Models

We evaluated different models using seven different metrics: Dice coefficient, intersection over union (IoU), precision (PRE), accuracy (ACC), sensitivity (SENS), specificity (SPEC), and the area under the curve (AUC).

$$Dice = \frac{2TP}{2TP + FN + FP} \quad (5)$$

$$IoU = \frac{TP}{TP + FN + FP} \quad (6)$$

$$PRE = \frac{TP}{TP + FP} \quad (7)$$

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

$$SENS = \frac{TP}{TP + FN} \quad (9)$$

$$SPEC = \frac{TN}{TN + FP} \quad (10)$$

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively. AUC stands for the area under the ROC curve.

The experiments were conducted on an HP workstation equipped with an Intel Core i7-10700 central processing unit and an NVIDIA GeForce RTX 3060 Ti graphics processing unit. All deep learning models were implemented and trained using the PyTorch framework (version 1.7.1). The codebase was developed in Python 3.8, utilizing libraries such as NumPy for numerical operations and Matplotlib for result visualization.

Results

Aorta Segmentation

The performance of different aorta segmentation models on the test dataset (which includes 8 TAAD, 8 TBAD, and 101 Non-AD) is shown in Table 2. Among the five segmentation models (3D U-Net, ResU-Net, SwinU-Netr, nnU-Net, and nnFormer), nnU-Net achieves the highest Dice coefficient (0.977), precision (0.972), sensitivity (0.983), and IoU (0.956). Table 2 also displays the aorta segmentation performance of different subject groups using nnU-Net. It demonstrates excellent performance, with a Dice greater than 0.970 across for all four groups. The Dice coefficient for Non-AD and TBAD is slightly higher than that for TAAD.

Figure 7 shows some segmentation results from the test dataset, including one case each of Non-AD, TAAD, and TBAD. The red regions represent the segmentation outcomes, with dissection visible in the TAAD and TBAD cases, providing a foundation for subsequent tasks.

Identification of Aortic Dissection

For the AD identification task, Table 3 presents the performance of various networks (VGG16, Inception-V3, ViT, and ResNet34) used in the feature extraction module, alongside our proposed method. Our method achieved the highest performance with an accuracy of 0.979, sensitivity of 0.982, specificity of 0.970, and AUC of 0.995. The ROC curve and confusion matrix are shown in Fig. 8. Notably, 280 out of 285 AD and 98 out of 101 Non-AD were correctly identified.

Table 2 Performance of five aorta segmentation networks and nnU-Net across different groups

Model	Dice	Precision	Sensitivity	IoU
3D U-Net (All)	0.575±0.111	0.548±0.133	0.628±0.118	0.412±0.106
ResU-Net (All)	0.507±0.091	0.390±0.097	0.764±0.140	0.344±0.084
SwinU-Netr (All)	0.914±0.032	0.875±0.040	0.958±0.037	0.843±0.052
nnFormer (All)	0.967±0.006	0.966±0.013	0.974±0.009	0.944±0.012
nnU-Net (All)	0.977±0.006	0.972±0.012	0.983±0.009	0.956±0.012
nnU-Net (Non-AD)	0.978±0.006	0.973±0.012	0.984±0.009	0.957±0.011
nnU-Net (AD)	0.973±0.006	0.967±0.012	0.979±0.006	0.948±0.010
nnU-Net (TAAD)	0.972±0.007	0.966±0.014	0.978±0.006	0.945±0.013
nnU-Net (TBAD)	0.974±0.003	0.969±0.009	0.980±0.006	0.950±0.007

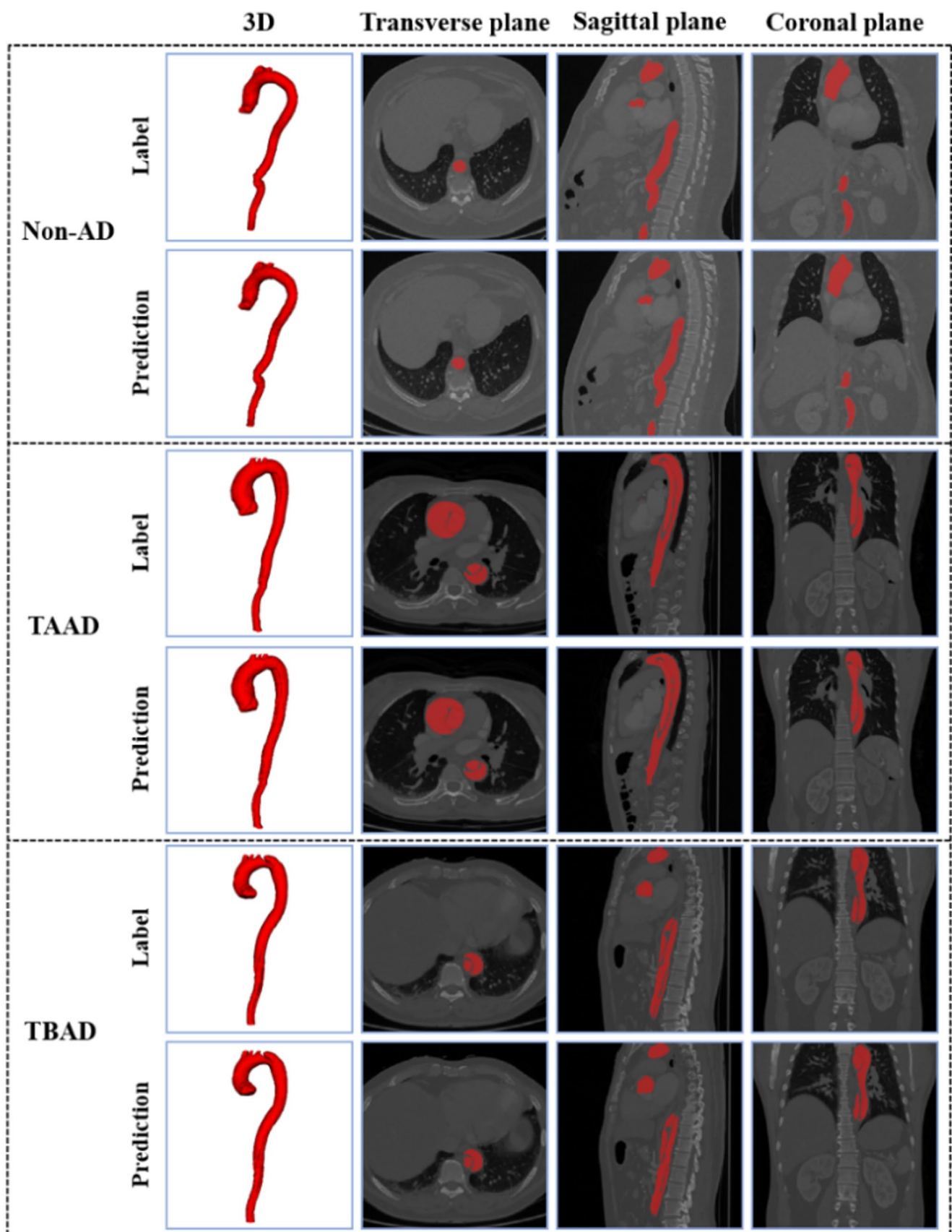
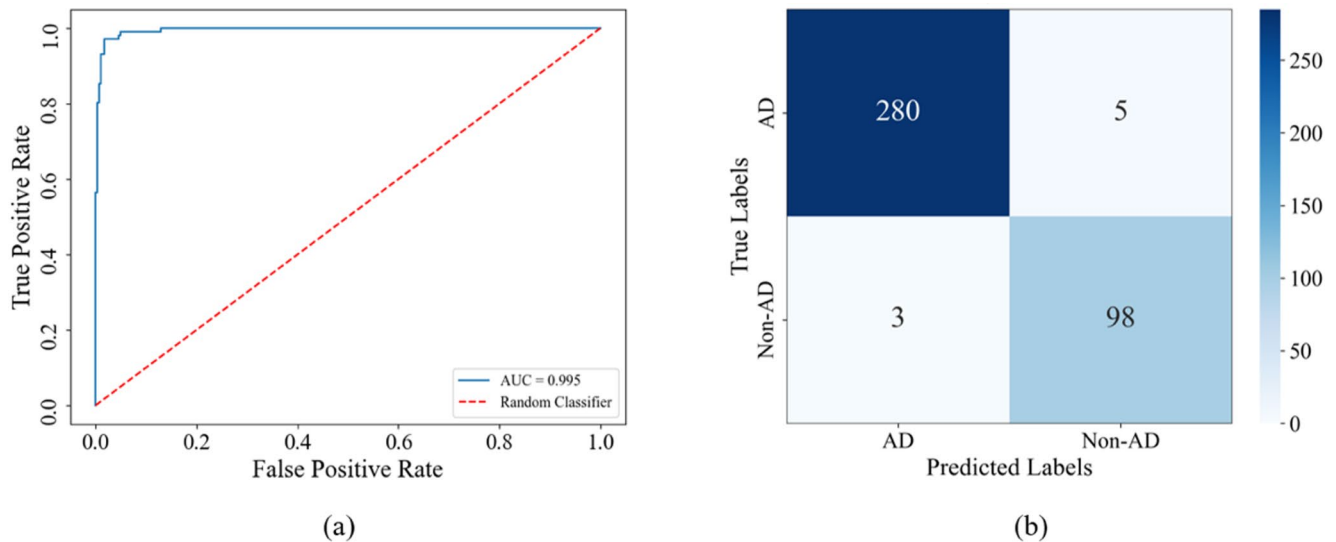


Fig. 7 Examples of aorta segmentation results in 3D view, transverse plane, sagittal plane, and coronal plane for Non-AD, TAAD, and TBAD

Table 3 Performance of multi-view classification models using different feature extraction networks for AD identification and Stanford subtyping

Task	Model	Accuracy	Sensitivity	Specificity	AUC
AD identification	VGG16	0.935	0.982	0.800	0.825
	Inception-V3	0.888	0.856	0.980	0.963
	ViT	0.738	1.000	0.000	0.848
	ResNet34	0.968	0.961	0.990	0.985
	Proposed method	0.979	0.982	0.970	0.995
Stanford subtyping	VGG16	0.760	0.535	0.966	0.758
	Inception-V3	0.975	0.964	0.986	0.990
	ViT	0.616	0.200	1.000	0.704
	ResNet34	0.975	0.964	0.987	0.989
	Proposed method	0.990	0.985	0.993	0.998

**Fig. 8** Classification performance of the proposed method for aortic dissection identification. (a) ROC curve; (b) Confusion matrix**Table 4** Performance of TL and FL segmentation using five different networks

Segmented part	Model	Dice	Precision	Sensitivity	IoU
True lumen	3D U-Net	0.582±0.135	0.437±0.146	0.935±0.009	0.423±0.136
	ResU-Net	0.405±0.114	0.520±0.184	0.339±0.089	0.261±0.093
	SwinU-Netr	0.532±0.101	0.518±0.139	0.569±0.090	0.370±0.096
	nnFormer	0.964±0.030	0.955±0.054	0.976±0.019	0.933±0.036
	nnU-Net	0.968±0.025	0.959±0.042	0.977±0.020	0.938±0.046
False lumen	3D U-Net	0.530±0.007	0.425±0.137	0.614±0.004	0.414±0.004
	ResU-Net	0.637±0.082	0.565±0.132	0.755±0.042	0.474±0.100
	SwinU-Netr	0.565±0.114	0.593±0.163	0.555±0.079	0.403±0.106
	nnFormer	0.961±0.028	0.971±0.033	0.953±0.045	0.935±0.051
	nnU-Net	0.971±0.033	0.979±0.039	0.965±0.036	0.946±0.057

* Bold indicates the best-performing one within the five models.

Segmentation of True Lumen and False Lumen

The performance of different models for TL/FL segmentation is shown in Table 4. Among the five segmentation models, nnU-Net achieved the best results on the test data, with a Dice of 0.968/0.971, precision of 0.959/0.979, sensitivity of 0.977/0.965, and IoU of 0.938/0.946. For TAAD data segmentation, nnU-Net reached a Dice of 0.959/0.966, while for TBAD data, it achieved a Dice of 0.976/0.976. nnU-Net

performed slightly better on the TBAD data compared to the TAAD data.

Figure 9 illustrates the TL/FL segmentation results for both TAAD and TBAD data, along with the generated dissected intima using morphological methods based on the segmentation results.

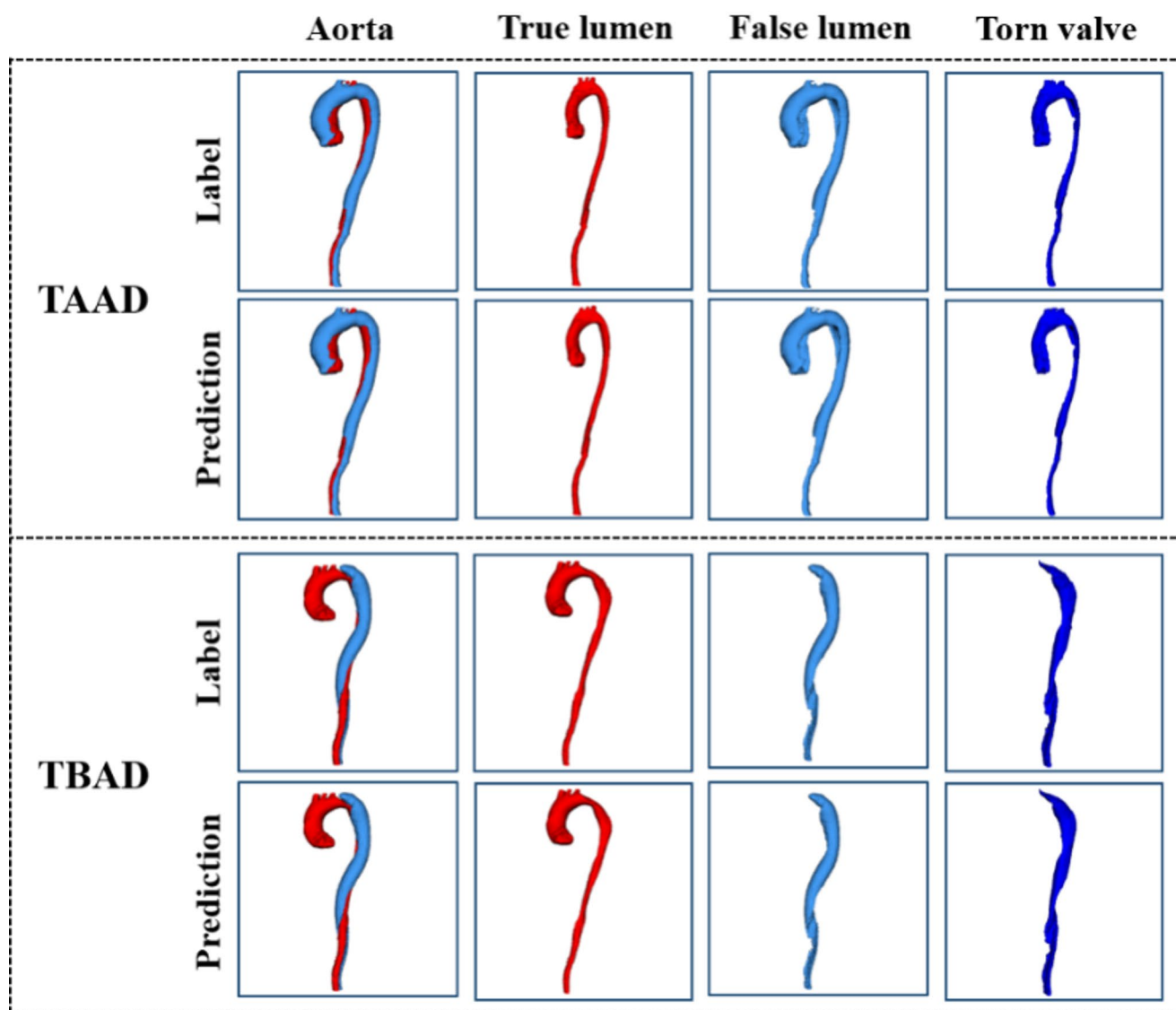


Fig. 9 Examples of segmentation results of the TL/FL, and morphologically generated torn intima in TAAD and TBAD

Stanford Subtyping

The lower part of Table 3 displays the Stanford subtyping performance of different networks (VGG16, Inception-V3, ViT, and ResNet34) used in the feature extraction module, along with our method. Among them, our method achieved the best performance, with an accuracy of 0.990, sensitivity of 0.985, specificity of 0.993, and AUC of 0.998.

Comparative Experiments

In the true and false lumen segmentation task, when the aortic segmentation results are not used as the ROI, the Dice coefficient of TL/FL segmentation decreases from 0.969 to 0.929. For the AD identification task (using ResNet34 with the SE block as the feature extraction network), replacing

multiple views with a single view (coronal plane) results in a performance drop, with the ACC decreasing to 0.952. Similarly, in the Stanford subtyping task (using ResNet34 with the SE block as the feature extraction network), using a single view (coronal plane) as input instead of multiple views leads to a decrease in classification performance, with the ACC dropping to 0.960.

External Validation

The segmentation performance of our model, trained on our collected dataset, is assessed using two external validation datasets. The calculation of each evaluation parameter covers the aortic branches and excludes the segmentation of false lumen thrombus. On the TBAD dataset [29], the Dice coefficients for the segmentation of Aorta, TL, and

FL are 0.907 ± 0.102 , 0.895 ± 0.124 , and 0.884 ± 0.163 for, respectively. Additionally, the IoU values are 0.843 ± 0.018 , 0.826 ± 0.156 , and 0.821 ± 0.193 . Table 5 presents a performance comparison between the method proposed in this study and the methods on the TBAD dataset. On the AVT dataset [33], our model achieves Dice, IoU, precision, and sensitivity of 0.749 ± 0.281 , 0.658 ± 0.268 , 0.878 ± 0.188 , and 0.719 ± 0.286 , respectively. These results demonstrate the promising generalization ability of our model.

Discussion

Aortic dissection is a type of aortic disease characterized by the tearing of the aortic intima, leading to the formation of TL/FL. Clinically, AD is classified into Stanford Type A and Stanford Type B based on the location of the intimal tear. Although the incidence of AD is low (approximately three cases per one hundred thousand individuals), the short-term mortality rate is high if left untreated [2, 3]. The development of effective treatment strategies for AD requires accurate classification based on the Stanford classification and formulation of treatment plans tailored to the lesion characteristics. Moreover, the high mortality rate in the short term without treatment highlights the urgent need for rapid diagnosis. In addition, precise treatment planning requires detailed 3D visualization of the aorta.

Compared to existing studies, our approach is pioneering. The pipeline proposed in this study uses the aortic segmentation task to guide the automatic identification of aortic dissection, followed by the segmentation of TL/FL. Based on the TL/FL segmentation results, Stanford subtype classification is then performed. Throughout the entire process, the pipeline operates fully automatically without requiring manual intervention. Furthermore, each module of this pipeline achieves performance equal to or better than existing methods. In the aorta segmentation module, the average Dice coefficient on the test set of 117 cases reached 0.977. In the AD identification module, the accuracy achieved 97.7% in both five-fold cross-validation-testing. For the true and false lumen segmentation module, the average Dice coefficients on the test set of 16 cases reached 0.968 and 0.971, respectively. In the Stanford subtyping module, the

accuracy achieved 99% in the five-fold cross-validation and test.

Aorta, True Lumen, and False Lumen Segmentation

In the aortic segmentation module, we trained the network using only AD data, excluding Non-AD data during the training process. This focused training strategy significantly enhances the network's attention on dissection structures, leading to improved segmentation performance compared to existing studies and designed comparative experiments.

In the TL/FL segmentation module, we used the aortic segmentation results as the ROI input for the network, rather than directly segmenting the TL/FL. This approach eliminates redundant information from the CT images, providing more focused guidance for TL/FL segmentation. As a result, it not only improves the network's operational efficiency but also enhances segmentation performance.

AD Identification and Stanford Subtyping

In the AD identification module, we performed patient-level classification using aortic projections, in contrast to the slice-level monitoring used in existing studies. The segmentation task was used to guide the classification, significantly improving both the accuracy and efficiency of aortic dissection detection. Notably, boundary extraction was applied to the aortic mask prior to projection to enhance dissection information. When combined with a multi-view strategy, this further boosted the performance of AD recognition.

In the Stanford subtyping module, the segmentation of TL/FL was used to guide classification. Additionally, combining maximum density projections of the true and false lumens with a multi-view approach enhanced their spatial features, further improving the performance of Stanford subtyping.

Comparison of our Method and the state-of-the-art Studies

The performance of our pipeline compared to existing methods for AD identification, segmentation, and subtyping is illustrated in Table 6. In the classification task, compared to the study by Huang et al. [31], Huang et al. [26], Dong et al. [27], and Hata et al. [28], our method directly obtains patient-level classification without relying on slice-level voting and empirically set thresholds. Furthermore, our study simultaneously covers data types including Non-AD, TAAD, and TBAD. Our method achieved an accuracy of 0.979 in the AD identification task, and 0.990 in the Stanford subtyping task, outperforming existing approaches. For the segmentation tasks of the aorta and TL/FL, previous studies by Yu et al. [21], Zhang et al. [22], Aghilinejad

Table 5 Performance comparison between the method proposed in this study and methods in the TBAD dataset

Segmentation	Method	Dice
Aorta	Proposed method	0.907 ± 0.102
	TBAD [23]	0.890 ± 0.003
TL	Proposed method	0.895 ± 0.124
	TBAD [23]	0.820 ± 0.008
FL	Proposed method	0.884 ± 0.163
	TBAD [23]	0.720 ± 0.020

Table 6 Performance of the proposed method and existing methods

Study	Task and dataset	Performance
Proposed method	1-Aorta segmentation 2- Identification of AD 3-Segmentation of TL/FL 4-Stanford subtyping 386 subjects (AD:285; Normal:101)	1-Dice=0.977 2-Accuracy=0.979; Sensitivity=0.982; Specificity=0.970; AUC=0.995 3-Dice of TL/FL=0.968/0.971 4-Accuracy=0.990; Sensitivity=0.985; Specificity=0.993; AUC=0.998
Yu et al., 2021 [21]	Segmentation and diameter of aorta and TL/FL with 3D CNN 139 TBAD patients	Dice of aorta=0.958 Dice of TL=0.961 Dice of FL=0.932
Zhang et al., 2023 [22]	Segmenting of TL/FL and aortic branches in TBAD with Semi-supervised method 100 TBAD patients	Dice of aorta=0.898 Dice of TL=0.910 Dice of FL=0.892
Aghilinejad et al., 2023 [11]	Segmenting the aorta with 2D CNN 19 TBAD patients	Dice of aorta=0.86
Feng et al. 2023 [12]	Segmentation of TL/FL and thrombus with two-step segmentation network. 167 postoperative TAAD patients	Dice of TL=0.967 Dice of FL=0.948
Xiang et al., 2023 [13]	Segmentation of aorta and TL/FL by two-step segmentation network 108 AD patients	Dice of aorta=0.932 Dice of TL=0.911 Dice of FL=0.884
Zhang et al., 2023 [19]	Segmentation of the aorta, thrombus, branch vessels, and TL/FL with a two-step segmentation network 133 postoperative TBAD patients	Dice of TL=0.962 Dice of FL=0.921
Huang et al., 2021 [31]	Identification of AD with first-step neural network and Stanford subtyping with second-step neural network 428 subjects (TAAD:79; TBAD:72; Normal:277)	Sensitivity of TAAD=0.955 Sensitivity of TBAD=0.793 Sensitivity of Normal=0.935 Specificity of TAAD=0.986 Specificity of TBAD=0.941 Specificity of Normal=0.941
Huang et al., 2024 [26]	Identification and classification of AD with two cascaded CNN networks at slice level 279 subjects (AD:173; Normal:106)\61,190 slices	Sensitivity=0.980 Specificity=0.976
Dong et al., 2024 [27]	Identification of AD with YOLOv5 coupled with CBAM module in slice level 480 subjects (AD:240; Normal:240)\9600 slices	Accuracy=0.915 AUC=0.938
Hata et al., 2021 [28]	Identification of AD with Xception in slice level and vote for patient level 170 subjects (AD:85; Normal:85)	Accuracy in patient-level=0.900 Sensitivity in patient-level=0.918 Specificity in patient-level=0.882

et al. [11], Feng et al. [12], and Zhang et al. [19] were all conducted on a single data category. In contrast, our study simultaneously covered Non-AD, TAAD, and TBAD types. Additionally, our method achieved Dice coefficients of 0.977 for the aorta, 0.968 for TL, and 0.971, without down-sampling the data.

Limitations and Future Studies

There are several limitations in our study. First, the use of data from a single center may limit the generalizability of the segmentation and classification models to other institutions or patient populations. Second, the relatively small size of the dataset could result in instability in the segmentation and classification models, necessitating further validation on larger datasets. Third, the dataset does not include data from other aortic diseases, such as aortic aneurysms or ruptures, which may reduce the specificity of the classification for these conditions. Lastly, the lack of cases with

insufficient contrast agent perfusion may result in poorer segmentation performance for this subset of data.

In future studies, we plan to expand the dataset to address some limitations. By incorporating data from multiple centers and diverse patient populations, we aim to enhance the performance of the segmentation and classification models across different settings. Additionally, we plan to include data from other aortic diseases, such as aortic aneurysms [44, 45] and ruptures [46, 47], to improve the specificity of the classification for various aortic pathologies. Moreover, efforts will be made to collect data on cases with insufficient contrast agent perfusion to improve the segmentation performance for these challenging scenarios. We also plan to incorporate new research strategies to deepen the model's understanding of images, such as semi-supervised learning [48–51]. Moreover, we aim to address issues related to insufficient perfusion and motion artifacts in CTA data through advanced techniques like Generative Adversarial Network [52] or other image generation methods [53, 54].

By addressing these limitations, we aim to further improve the accuracy and reliability of our deep-learning pipeline for AD identification and characterization.

Conclusion

In summary, we proposed a fully automated pipeline for detecting AD, performing Stanford subtyping, and segmenting the TL/FL from CTA images. By adding modules for aorta segmentation and employing multi-view input in the AD identification and Stanford subtyping modules, we significantly improved the classification performance. Due to its emulation of radiologists' diagnostic procedures, the pipeline is user-friendly for radiologists and ensures interpretability. The proposed pipeline has the potential to expedite diagnosis and provide valuable insights for treatment planning.

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Declarations

Ethical Approval The study was in accordance with the ethical standards of the committee of the General Hospital of Northern Theater Command and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

Informed Consent Informed consent was waived due to the retrospective nature of the study, which made obtaining consent impractical.

Conflict of Interest The authors declare that they have no competing interests.

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